

Jest Unit Test Coverage Report

tnn-english-next | Times Now News (English)

Report date: 2026-07-14 | Jest coverage + __tests__ inventory

1. Executive Summary

This report inventories the Jest unit test suite for the TNN English Next.js application: what is covered, coverage percentages by area, component-level unit testing depth, and the classes of bugs these tests are designed to catch.

Test files	Static it/test cases	Rewrite rules tested	Overall line coverage
324	3,277	203	89.3%
Statements	Functions	Branches	Lines covered
88.2%	75.9%	66.3%	26,958/30,195

Suite size: 324 *.test.js files under __tests__/. About 3,277 explicitly written it()/test() cases. URL rewrite suites add factory-generated cases for all 203 production rewrite rules (~1218+ additional dynamic tests with positive, negative, and variant path checks). Environment: jest-environment-jsdom with Testing Library; Next.js router/head/script and cheerio mocked in setup.

2. Coverage by Source Area

Based on coverage/coverage-summary.json (jest --coverage collectors for components, pages, utils, hooks, helpers, constants, middleware, routes).

Area	Files	Lines	Branches	Functions	Lines cov
components	756	88.3%	64.9%	78.6%	16001/18128
pages	82	89.2%	62.1%	51.5%	7252/8131
utils	36	95.8%	82.0%	98.5%	2167/2261
helpers	60	90.7%	73.4%	87.2%	1176/1296
constants	5	98.6%	74.4%	100.0%	212/215
hooks	6	90.8%	73.3%	90.0%	139/153
middleware	1	100.0%	100.0%	100.0%	10/10
routes	1	100.0%	0.0%	0.0%	1/1

3. Test Inventory by Category

Folder	Files	Cases	What it exercises
components	140	1539	React UI (render, props, variants)
pages	62	865	Page shells, GSSP, pages/api helpers
utils	60	630	Pure helpers (dates, URLs, ads)
routes	33	~1218+ dyn	URL rewrites (dynamic factory suites)
helpers	16	142	SEO schemas, analytics helpers
constants	6	48	Exports and URL mapping
hooks	5	19	Custom React hooks
middleware	1	4	Next middleware

Folder	Files	Cases	What it exercises
api	1	17	API-related

4. Component Unit Testing Breakdown

Component tests use React Testing Library in jsdom. Coverage and case volume concentrate on shared UI (common), platform shells (desktop/mobile/AMP), and page compositions. Many suites are mega/deep/batch files that push branch coverage on large presentational components.

4.1 Test cases by component domain

Domain	Files	Cases	Focus
common	46	616	Shared widgets, forms, players, ads, carousels
platforms	20	340	Desktop/mobile/AMP header/footer/nav shells
amp	17	178	AMP page components and live blog/video
desktop	13	107	Desktop home, article, sports, live blog
mobile	11	62	Mobile article/home/video/live blog
layout	7	53	Layout shells and component map smoke
pages	3	26	Page-level composition shells
election	2	23	Election widgets/tally UI
liveBlog	2	20	Live blog UI
sections	3	20	Home/section layout sections
widgets	2	20	Games, premium, bottom text widgets
prime	2	15	Prime article experience
zoom	2	15	Zoom listing/detail UI
newArticleShow	2	14	Article show SEO + multi-platform
video	2	7	Domain-specific UI coverage
weather	1	6	Domain-specific UI coverage
listing	1	5	Domain-specific UI coverage
babyNames	1	4	Domain-specific UI coverage
photoGallery	1	3	Domain-specific UI coverage
recipe	1	3	Domain-specific UI coverage
home	1	2	Domain-specific UI coverage

4.2 Source coverage by components/* subfolder

Subfolder	Files	Line %	Branch %	Lines
platforms	441	87.3%	64.0%	9460/10834
common	288	89.0%	65.7%	5889/6619
layout	12	98.4%	83.3%	428/435
widgets	6	92.6%	70.0%	112/121
election	7	93.7%	75.8%	74/79
recipe	1	97.2%	86.8%	35/36
amp	1	75.0%	71.4%	3/4

Overall components line coverage: 88.3% across 756 collected files. Branch coverage (~65%) is the weakest metric: optional UI paths (flags, rare props, error UI) are less exercised than happy paths.

5. What Each Test Layer Covers

5.1 Components (140 files, ~1,539 cases)

- Smoke/render: components mount without throwing for typical props fixtures.
- Conditional UI: mobile vs desktop vs AMP, feature flags, empty/missing data.
- Content presentation: headlines, images, sticky nav, live blog ticks, video/players.
- Widget composition: home widgets, listing cards, election, weather, IFSC, recipes, zoom.
- Layout mapping: layout shells and component-map registry smoke tests.

5.2 Pages and GSSP (62 files, ~865 cases)

- getServerSideProps for home, article show (new/old), AMP variants, live blog, sports, zoom, recipe, podcast, board results, silo, info, vertical pages.
- pages/api helpers: Redis/flat-cat resolution, widget data, header/footer, external fetches, layout utils, de-detail/photo.
- Page shell batches: root and AMP page modules export/render correctly with mocked data.
- Bug focus: bad props shapes, missing API payloads, redirect/notFound, Redis miss fallbacks.

5.3 Utils (60 files, ~630 cases)

- Pure functions: date formatting, HTML decode, link/URL builders, cookies, isMobile, logging.
- Networking: makeRequest agents/errors, server utils, loadAdsConfig branches/pages.
- Election SSR: tally merge/inject, error paths (some election tests ignored in jest.config).
- Media/layout: NextImage preload, ampPathUtils, parseLayoutData, zoom listing payload, board result URLs.

5.4 Routes / Rewrites (33 files)

- All 203 rewrite rules in routes/tnn_rewrites.js are catalogued and assigned to category suites.
- Per rule: positive source match, destination declared, param capture, rewrite chain resolution, path variants, negative malformed paths.
- Also: duplicate sources, shadowed rules, blocked /404 rules, priority/conflict analysis.
- Bug focus: broken regexes, wrong destination pages, param loss, earlier rules shadowing later ones, 404 misclassification.

5.5 Helpers - SEO and Analytics (16 files, ~142 cases)

- JSON-LD / schema builders: NewsArticle, FAQ, Video, ImageGallery, Breadcrumb, Organization, WebSite.
- Meta helpers: DynamicMetasLinks, Analytics injection.
- Analytics: gaUtils, Chartbeat categories, GRX, Colombia DMP, Marfeel, scheduleTask.

5.6 Hooks, Constants, Middleware

- Hooks: useWidgetImpression, useArticleScrollTracking, useBottomStickyNavScroll, usePassiveRafScroll, breaking-news check.
- Constants: exhaustive export checks, urlMapping / findUrlMappingRedirect.
- Middleware: full line/branch coverage of middleware.js.

6. Types of Bugs These Tests Can Find

These are unit/integration-style Node+jsdom tests. They catch logic and contract bugs early; they do not replace browser E2E, visual regression, or real Redis/CDN/API soak tests.

Bug class	Catchability	Evidence in suite
Null/empty props crashes	High	~403 titles mention null/empty/missing
Wrong SSR props / GSSP failures	High	GSSP suites: notFound, redirects, shapes
Broken URL rewrites / redirects	High	203 rules: match, params, shadowing, 404
Incorrect SEO JSON-LD / metas	High	Schema builders + article SEO suites
Branch/flag UI regressions	Med-High	Desktop/mobile/AMP and flag branches
Ads config / tracking breaks	Medium	loadAdsConfig + gaUtils / impressions
Date / HTML / URL formatting	Medium	dateUtils, htmlDecoder, linkUtils
Click/handler regressions	Medium	~138 interaction-oriented cases
Election/sports live data merge	Medium	electionTallySsr merge/error suites
Export/rename breakage	Medium	constants/helpers exhaustive exports
Visual/CSS layout bugs	Low	jsdom cannot assert real layout/paint
Real network/Redis race bugs	Low	Most I/O is mocked
Cross-browser quirks	Low	Single jsdom environment
Performance / memory leaks	Low	Not load/perf tested

6.1 Strengths

- Broad source coverage (~89% lines) across the main app surface.
- Strong pure-logic coverage in utils (~96% lines, ~98% functions).
- Systematic rewrite regression protection across the full route catalog.
- Heavy SSR/GSSP and pages/api helper testing for news page data contracts.
- SEO schema unit tests reduce structured-data regressions.

6.2 Gaps / residual risk

- Branch coverage overall is 66.3% - many else-paths and rare conditions untested.
- Pages function coverage is only ~51% - some page functions lightly touched.
- Two tests are ignored in jest.config (elections rewrites, electionPresiteFlags).
- Lower-covered component files still include players, drawers, nav, old-article paragraphs, IFSC forms (see section 7).
- No guarantee of production visual fidelity or live third-party script behavior.

7. Lower-Covered Hotspots (priority gaps)

Collected source files with >=25 executable lines and line coverage below 75% (candidates for follow-up unit tests):

Line %	File	Area	Covered
14.1%	OldArticleLeadPlayer.js	components	11/78
46.1%	Drawer.js	components	12/26
52.8%	OldArticleParagraph.js	components	19/36
58.8%	index.js	components	30/51
60.0%	DrupalPhotoDetails.js	components	15/25
60.0%	ShareIcon.js	components	15/25
61.3%	MovieReviewParagraph.js	components	38/62
62.2%	DrupalDynamicMetasLinks.js	seo	23/37
63.1%	MovieReviewParagraph.js	components	24/38
64.0%	VideoPlayer.js	components	16/25
66.7%	LiveTvWidget.js	components	24/36
68.2%	PodcastShow.js	components	45/66
69.8%	HomeWidget1.js	components	30/43
71.0%	Paragraph.js	components	22/31
71.4%	Paragraph.js	components	25/35
71.8%	MovingCube.js	components	28/39
72.3%	SportsHPWidget.js	components	34/47
72.5%	Paragraph.js	components	29/40
73.0%	NavMenu.js	components	46/63
73.8%	SlickWrapper.js	components	31/42
74.4%	Header.js	components	32/43
74.5%	EmbedLiveBlog.js	components	35/47
74.6%	ShowCoverVideo.js	components	44/59

8. Notable GSSP Page Suites

Dedicated getServerSideProps tests under __tests__/pages/gssp/:

GSSP suite	GSSP suite
articleShow	boardResults
home	homeAmp
info	legacyAmpPages
liveBlog	newArticleShow
newArticleShowAmp	oldArticleShow
oldPhotoDetail	pagesGsspMegaBatch
pagesGsspMegaBatch2	pagesRegistry
podcastShow	recipeShow
rootPages	silo
sports	verticalAmpPages
verticalPages	zoom

9. Tooling Snapshot

- Runner: Jest 30.x via next/jest; scripts: npm test / test:coverage / test:ci
- DOM: jest-environment-jsdom + @testing-library/jest-dom

- Setup: `__tests__/setup/jest.setup.js` (fetch mock, matchMedia, next/router|head|script, cheerio)
- Coverage reporters: text, text-summary, lcov, json-summary
- collectCoverageFrom: components, pages, utils, hooks, helpers, constants, middleware, routes
- Timeout: 20s; forceExit: true; some election tests pathIgnored

10. Verdict

The project has a large, coverage-driven unit suite (324 files; ~3,277 static cases plus factory-generated rewrite suites for 203 rules) achieving ~89% line and ~88% statement coverage. It is strongest at preventing SSR contract breaks, rewrite regressions, SEO schema mistakes, and component crash/regression on missing data. It is weaker at exhaustive branch paths, real browser visuals, and live I/O. Treating branch coverage (66%) and the section 7 hotspots as the next quality investment would yield the best residual-risk reduction.